Discourse vs. Reality: Correlating Political Manifestos with Economic Indicators via Topic Modeling

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Abstract

Against the backdrop of France's transition to structural mass unemployment between 1981 and 1993, this study tries to examine the territorial anchoring of political campaigns. By joining the *Archelec* corpus of legislative electoral manifestos with localized macroeconomic data from the National Institute of Statistics and Economic Studies (INSEE), I investigate whether candidates adapt their rhetoric to specific departmental economic conditions or strictly adhere to centralized national party lines. To accurately measure the discursive space allocated to the economic crisis, I implement a comparative Natural Language Processing (NLP) framework. This study contrasts a deterministic dictionary-based baseline with an advanced Zero-Shot semantic pipeline, leveraging CamemBERT embeddings and a chunk-level max-pooling strategy within the BERTopic framework. Methodologically, the results expose the limitations of "bag-of-words" techniques, which systematically overestimate thematic intensity by ignoring syntactic context and polysemy. Empirically, both NLP pipelines reveal a persistent lack of spatial correlation between the preponderance of unemployment discourse and actual departmental unemployment rates. This statistical disconnect provides quantitative evidence for the "nationalization" of French political communication. It sheds light on the fact that local economic discourse operated primarily as a top-down rhetorical posture rather than a targeted territorial diagnostic.

1 Introduction

Between 1981 and 1993, France transitioned from post-war growth to structural mass unemployment, culminating in the severe 1993 recession. This economic crisis reshaped the national political landscape and altered campaign strategies. Legislative elections in the French Fifth Republic present a paradox: deputies are elected in local constituencies but campaign under the banners of centralized national parties. This dynamic raises a critical question regarding political communication: to what extent does local political discourse actually reflect local economic realities?

To investigate the territorial anchoring of political campaigns, this study uses Natural Language Processing (NLP) to analyze the Archelec corpus, a digitization of electoral manifestos from the 1981, 1988, and 1993 elections. By quantifying the discursive space allocated to unemployment and cross-referencing these metrics with departmental unemployment rates from the National Institute of Statistics and Economic Studies (INSEE), this research estimates the "nationalization" of French campaigns. Furthermore, a comparative NLP framework is implemented to distinguish between the superficial use of political buzzwords and contextually grounded argumentation.

2 Related Work

First, from a theoretical standpoint, this study intersects with classical political science literature regarding the spatial dynamics of electoral behavior. Traditional comparative politics, notably

Caramani's work on the "nationalization of politics" Caramani (2004), advocates that modern European elections have progressively shed their localized specificities in favor of nationwide political cleavages. Furthermore, the sociology of political communication, such as Le Bart's analysis of political discourse Le Bart (1998), highlights the highly constrained and standardized nature of electoral manifestos.

On the technical hand, large-scale initiatives like the Comparative Manifestos Project (CMP) traditionally relied on manual human coding to measure political issue salience Budge et al. (2001). While highly accurate, manual coding is resource-intensive and impractical for massive corpora like the Archelec database, necessitating the adoption of automated text analysis.

Early automated approaches in political science primarily utilized "bag-of-words" methodologies. Dictionary-based keyword matching and scaling algorithms like Wordscores Laver et al. (2003) positioned political actors based purely on word frequencies. Additionally, unsupervised topic models like Latent Dirichlet Allocation (LDA) Blei et al. (2003) became standard for uncovering latent thematic structures. However, these lexical methods ignore syntactic context. They struggle with polysemy and frequently trigger false positives when political buzzwords are used superficially.

Recently, the NLP landscape has shifted toward deep learning and semantic representation. Transformer-based architectures, such as the French-specific CamemBERT model Martin et al. (2020), generate context-aware word embeddings. Frameworks like BERTopic Grootendorst (2022) manages these embeddings to enable precise Zero-Shot classification, clustering documents by semantic proximity rather than surface vocabulary. While frequently applied to modern social media, their application to historical archives remains underexplored. This study bridges that gap by using BERTopic to isolate economic discourse from mere rhetoric, providing a metric for comparison against INSEE data.

3 Data and Preprocessing

To construct a reliable comparative analysis, this study aligns a textual corpus of political manifestos with a macroeconomic statistical dataset. Both underwent standardized preprocessing.

3.1 The Archelec Corpus

The primary textual data derives from the *Archelec* Po (2024) project. This study focuses on the legislative elections of 1981, 1988, and 1993, a period marked by the entrenchment of mass unemployment.

To preserve semantic integrity, the preprocessing pipeline first segmented documents into distinct paragraphs, preventing the dilution of localized economic arguments within multi-thematic documents. Short, non-argumentative elements (e.g., formatting artifacts, biographical bullet points) were systematically filtered out. The remaining text was cleaned of residual OCR artifacts to ensure clean inputs for the embedding models. Notably, for the 1981 elections, macroeconomic data from 1982 is utilized as the closest reliable temporal proxy available in the INSEE historical archives.

Election Year	INSEE Data Year	Documents
1981	1982	3,181
1988	1988	3,620
1993	1993	5,936

Table 1: Summary of the Archelec corpus and INSEE macroeconomic data.

Figure 1 depicts the Top 20 French departments ranked by candidate frequency, quantified by distinct manifestos, for this election. This visualization effectively identifies departments experiencing highly concentrated electoral competition. Paris (75) exhibits the highest candidate density, followed by Nord (59) and Seine-Saint-Denis (93).

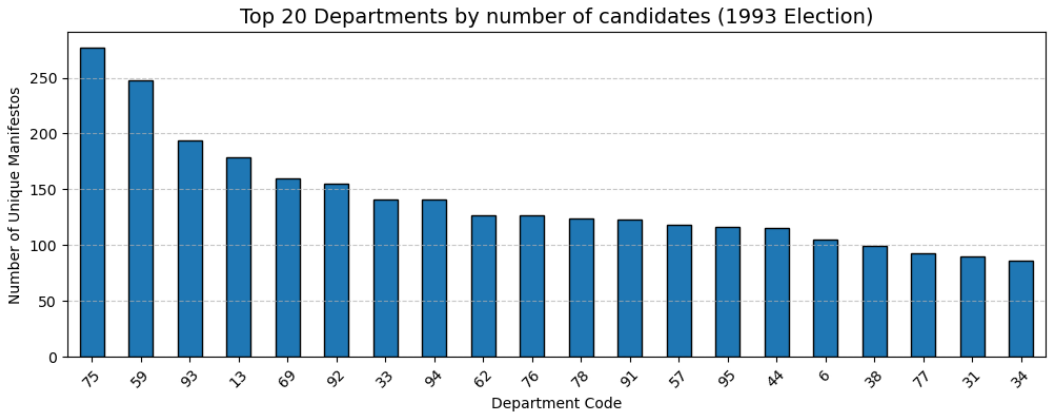

Figure 1: Ranking of top 20 French departments by number of manifestos in the 1993 legislative election.

3.2 INSEE Unemployment Data

To establish a local macroeconomic ground truth, I utilized historical departmental unemployment statistics provided by INSEE INSEE (2024). To synchronize this quarterly data with the electoral timelines, quarterly rates were averaged to generate a single annual macroeconomic indicator per department for each election year. As illustrated in Figure 2, mapping these indicators visually establishes the historical reality of the crisis. It highlights stark regional disparities providing the geographic baseline required to test the territorial anchoring of the political manifestos. For instance, the severe economic degradation in the industrialized North and East can be compared to more stable rural departments.

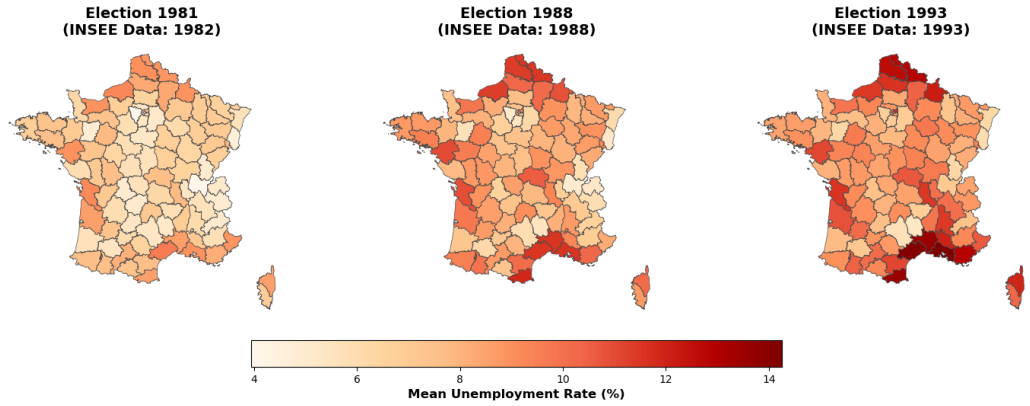

Figure 2: Evolution of average departmental unemployment rates (INSEE) across the studied years.

This pipeline yielded a structured dataset where the discursive intensity of candidates could be directly compared with the localized economic severity of their respective constituencies.

4 Methodology

To accurately capture macroeconomic reality within the political text, I structured the NLP analysis as a methodological funnel.

4.1 Baseline Exploration: Unsupervised and Seeded LDA

I initially deployed a standard Latent Dirichlet Allocation (LDA) model using a `CountVectorizer` to explore the corpus's latent thematic structure. Unsupervised LDA revealed a high density of

"electoral noise." Topics were dominated by administrative words (e.g., *"maire"*, *"circonscription"*) rather than substantive policy. Consequently, unemployment discourse appeared too semantically fragmented to form a cohesive topic, diluting economic terms across heterogeneous clusters.

In standard LDA Blei et al. (2003), each topic k is characterized by a multinomial distribution over the vocabulary V, denoted ϕ_k, drawn from a symmetric Dirichlet prior:

$$\phi_k \sim \text{Dir}(\beta \mathbf{1}_V) \tag{1}$$

Because the scalar smoothing parameter β is applied uniformly, the model possesses no initial lexical preference, causing the observed fragmentation.

The Seeded LDA approach Jagarlamudi et al. (2012) breaks this symmetry using asymmetric lexical priors. For the target unemployment topic k^*, I defined seed words $S_{k^*} = \{\textit{chômage}, \textit{emploi}, \textit{licenciement}\}$. The Dirichlet prior is modified into a parameter vector $\boldsymbol{\tau}_{k^*}$:

$$\tau_{k^*,w} = \beta + \lambda I(w \in S_{k^*}) \tag{2}$$

Where β is the baseline noise, $\lambda > 0$ dictates the lexical bias strength, and $I(\cdot)$ is the indicator function for the seed set. The word distribution is then sampled as:

$$\phi_{k^*} \sim \text{Dir}(\boldsymbol{\tau}_{k^*}) \tag{3}$$

This λ bias artificially shifts probability mass toward the seeds, forcing the algorithm to agglomerate fragmented economic discourse into a cohesive policy topic. However, Seeded LDA still relies on discrete word counts. To capture context without dilution, a transition to dense semantic vector representations was necessary.

4.2 BERTopic Framework

To distinguish the actual semantic intent, differentiating "work" as a social value from "lack of work" as a crisis, I implemented a Zero-Shot classification pipeline managing CamemBERT embeddings Martin et al. (2020) within BERTopic Grootendorst (2022).

This approach projects both the text and a predefined semantic anchor z into a shared vector space R^d. With embedding function $E(\cdot)$, the target concept is $\mathbf{v}_z = E(z)$.

To prevent the dilution of strong localized arguments within longer paragraphs, I applied the following strategy. A paragraph P is segmented into smaller chunks $C = \{c_1, \ldots, c_n\}$, embedded as $\mathbf{v}_{c_i} = E(c_i)$. The semantic alignment between a chunk and the target concept is computed via cosine similarity:

$$\text{sim}(c_i, z) = \frac{\mathbf{v}_{c_i} \cdot \mathbf{v}_z}{\|\mathbf{v}_{c_i}\| \|\mathbf{v}_z\|} \tag{4}$$

To aggregate these similarities, I applied a max-pooling operation. The final discursive intensity score S for paragraph P is the maximum chunk similarity:

$$S(P) = \max_{c_i \in C} (\text{sim}(c_i, z)) \tag{5}$$

Utilizing the $\max$ operator rather than a simple average ensures that paragraphs dedicating even a single sentence to unemployment are flagged. This mathematically prevents the dilution caused when economic arguments are embedded within multi-thematic paragraphs.

4.3 Dictionary-Based Lexical Scoring

To benchmark the semantic model, I also established a deterministic dictionary-based baseline relying on strict token matching. Let lexicon $L = \{\textit{chômage}\}$. A paragraph P is treated as a sequence of N tokens. Target word occurrence is:

$$I(w_i \in L) = \begin{cases} 1 & \text{if } w_i \in L \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

A paragraph is flagged as relevant ($y_P = 1$) if it contains at least one lexicon token:

$$y_P = I\left(\sum_{i=1}^{N} I(w_i \in L) > 0\right) \tag{7}$$

To align with INSEE data, paragraph scores are aggregated geographically. For department d with paragraphs M_d, the regional discursive intensity $S_{lex}(d)$ is:

$$S_{lex}(d) = \frac{1}{|M_d|} \sum_{P \in M_d} y_P \tag{8}$$

While computationally efficient, this formulation reveals the limits of lexical proxies. Because the inner summation ignores syntactic structure, y_P systematically triggers false positives on superficial buzzwords, inflating the departmental score.

5 Experiments and Results

5.1 Qualitative Validation of the Semantic Filter

To qualitatively validate the precision of the Zero-Shot pipeline, a manual review of the highest-scoring paragraphs was conducted. Extracts from the 1993 elections—the historical peak of the targeted economic crisis—illustrate the model's capacity to isolate substantive policy argumentation over mere keyword occurrences.

Paragraphs achieving the maximum semantic similarity scores consistently frame unemployment as the central electoral battleground. The model successfully captures direct diagnostics, flagging manifestos that declare: *« Un seul vrai débat électoral mérite d'être mené aujourd'hui : celui des mesures à prendre contre le chômage. Nous payons tous en effet au prix fort les conséquences de ce gâchis économique et humain. »* This demonstrates the algorithm's ability to identify the framing of unemployment as a societal waste rather than a mere statistical fact.

Similarly, the pipeline accurately identifies opposition rhetoric calling for structural macro-management, capturing statements such as: *« [La nouvelle Majorité] va oeuvrer pour redresser notre économie, combattre ce fléau qu'est le chômage. »* Here, the semantic model correctly weighs the gravity of the term *fléau*.

Furthermore, the embeddings prove capable of capturing complex, multi-variable policy linkages without explicitly relying on the target word itself. For instance, the model highly scored paragraphs proposing systemic educational solutions for youth job creation: *« réformer avec courage notre système éducatif. Il faut former les jeunes pour les emplois de demain et non pas ceux d'hier. [...] réhabiliter l'enseignement technique et professionnel. »* This qualitative review confirms that the BERTopic scores do not represent a simple frequency count, but a highly accurate measure of concentrated political intent.

5.2 Longitudinal Evolution of Discourse

To measure the evolution of economic anxiety, the national average of discursive intensity dedicated to unemployment was calculated per election year (Table 2).

Year	Lexical (%)	BERTopic (%)
1981	5.30	2.80
1988	7.93	2.95
1993	10.35	4.57

Table 2: National average of discursive space dedicated to unemployment.

Both methodologies capture the same trajectory: a steady rise in unemployment salience peaking during the 1993 recession. However, the lexical approach estimates more than double the discursive weight of the semantic model (10.35% vs. 4.57% in 1993). This inflation confirms that the dictionary approach acts as a noisy net.

Conversely, BERTopic provides a conservative metric. Qualitative review validates that the model successfully isolated explicit economic rhetoric. The 4.57% recorded in 1993 represents a highly accurate measure of true political argumentation.

5.3 Methodological Agreement: Lexical vs. Semantic

Before evaluating the political discourse against macroeconomic realities, I assessed the internal agreement between the two NLP pipelines. As visualized in Figure 3, comparing the departmental scores generated by both methods across the three electoral cycles reveals a moderate correlation hovering around $r \approx 0.60$.

This variance graphically illustrates the structural differences formalized in the methodology. The scatter plots demonstrate that while both models detect the same trend, the lexical pipeline systematically inflates the estimated intensity by capturing both substantive arguments and passing rhetoric. On the other hand, BERTopic tightens the distribution by discarding diluted mentions. Both detect the rising economic anxiety, but they measure its intensity at fundamentally different scales of precision.

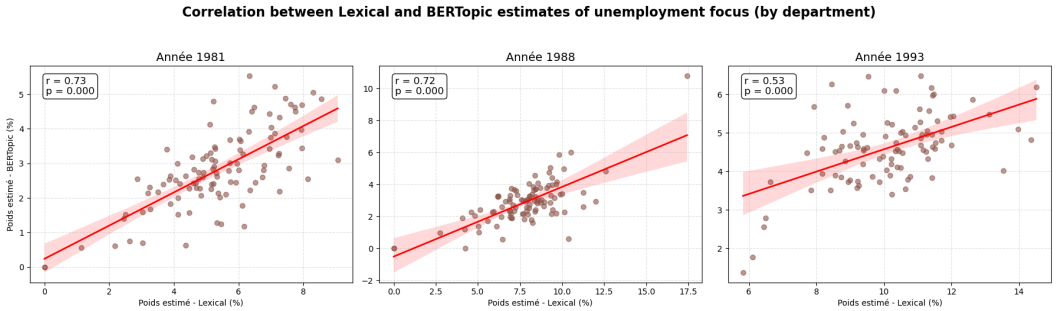

Figure 3: Correlation between Lexical and BERTopic discursive scores ($1981 - 1993$).

5.4 Discourse vs. Economic Reality (INSEE Correlations)

To rigorously test the primary hypothesis regarding the territorial anchoring of political campaigns, the text-derived discursive intensity metrics were regressed against the macroeconomic ground truth provided by the departmental INSEE unemployment rates. I conducted a series of regressions and computed correlation coefficients for each electoral year, comparing the outputs of both the lexical baseline and the BERTopic pipeline.

However, these statistical outputs must be interpreted with strict methodological caution. First, the inherently unsupervised nature of the NLP pipeline warrants prudence. Because the semantic models operate without manually labeled ground-truth data, the resulting discursive intensity scores remain algorithmic proxies rather than precise econometric variables. Regressing unsupervised, text-derived metrics against hard macroeconomic statistics inevitably introduces an irreducible margin of interpretive noise.

Second, the empirical outputs themselves dictate caution. The results reveal a persistent lack of statistical alignment, characterized by near-zero correlation coefficients (r) and consistently high p-values across all three election cycles. Because these p-values remain well above standard significance thresholds (e.g., $p > 0.05$), we cannot reject the null hypothesis of independence between the variables. This strict absence of correlation is visually confirmed by the flat trend lines in the scatter plots of Figure 4. The variance in political rhetoric simply cannot be explained by the spatial heterogeneity of the economic crisis. Candidates campaigning in severely depressed departments did not allocate a statistically larger proportion of their manifestos to the unemployment crisis than those running in relatively stable constituencies.

Crucially, the transition from the noisy dictionary-based approach to the high-fidelity semantic model did not alter this outcome. One might initially hypothesize that the lack of correlation in the lexical baseline was a mere statistical artifact, caused by the algorithm's inability to filter out superficial political buzzwords. However, even when utilizing BERTopic's, the geographic decoupling remains absolute. The semantic filtering drastically reduced the noise in the data, yet uncovered no hidden spatial correlation. This methodological convergence definitively confirms that the absence of alignment is not an NLP failure, but a profound sociological reality of the French electoral system.

Figure 4: INSEE Unemployment Rate vs. Predicted Discursive Intensity.

5.5 Discourse vs. Economic Reality (Spatial Distribution)

The cartographic projection of these metrics, presented in Figure 5, provides a visual confirmation of the statistical disconnect observed in the regression analysis. By juxtaposing the two datasets, a spatial contrast emerges. The actual INSEE unemployment rates (left panel) display sharp regional disparities. This ground-truth macroeconomic data clearly delineates the uneven geography of the crisis.

Conversely, the NLP-generated spatial distribution derived from the *Archelec* manifestos (right panel) reveals an homogenized landscape of political rhetoric. The discursive intensity surrounding unemployment completely fails to cluster around the actual economic disaster zones. Instead of mapping the macroeconomic heterogeneity of the crisis, the political discourse appears uniformly saturated across the entire metropolitan territory. Whether a department was experiencing severe unemployment or relative economic stability, the electoral space dedicated to the issue remained constant.

This visual evidence strongly reinforces the core quantitative conclusion of this study. The thematic structure of local electoral manifestos is detached from geographic realities. The maps visually demonstrate that unemployment was not treated as a localized policy challenge requiring territorial solutions. Rather, it functioned as an omnipresent national narrative which is divorced from the actual macroeconomic health of the populations they sought to represent.

6 Discussion and Limitations

6.1 The Nationalization of Discourse and Rhetorical Noise

The spatial disconnect observed in the data fundamentally challenges the rational-choice assumption that local candidates strictly tailor their campaigns to the specific socio-economic grievances of their constituencies. Instead, the findings provide quantitative evidence for the structural "nationalization" of French political campaigns during the Fifth Republic. Within this framework, the local electoral manifesto functions less as a territorial diagnostic and more as a top-down communicative vehicle dictated by national party headquarters. Candidates deploy a highly homogenized rhetoric. This suggests that legislative elections during this period operated largely as national referendums, where the candidate serves as a proxy for a centralized ideological platform rather than an independent local problem-solver.

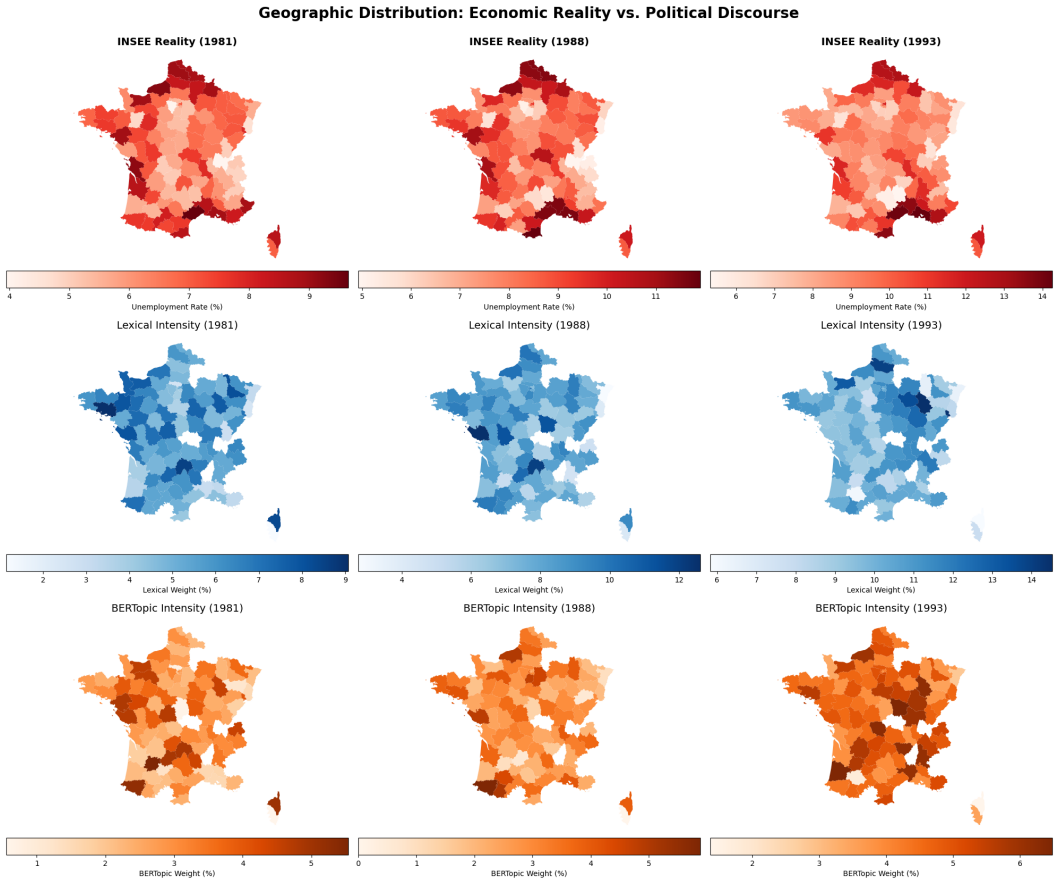

Figure 5: Geographical comparison: Actual INSEE unemployment rates (left panel) versus BERTopic predicted discursive intensity (right panel) across the 1981, 1988, and 1993 elections.

Consequently, the discourse surrounding unemployment between 1981 and 1993 did not operate as a localized policy response. Rather, it functioned as a mandatory rhetorical figure permeating the entire political spectrum. As structural mass unemployment became deeply entrenched in the French collective psyche following the economic shocks of the 1970s, acknowledging the crisis transformed into a strict prerequisite for political credibility. The data reveals that candidates engaged in this discourse evoked the economic crisis as a matter of national posture and obligatory empathy, entirely decoupled from the objective demographic severity of their immediate electoral environment.

These findings provide quantitative validation for Caramani's Caramani (2004) theoretical framework regarding the nationalization of politics. Within this framework, the local electoral manifesto functions less as a territorial diagnostic and more as a top-down communicative vehicle dictated by national party headquarters. Consistent with Le Bart's Le Bart (1998) observations on the standardization of political language, candidates deploy a highly homogenized rhetoric.

6.2 Methodological Contributions

The research also demonstrates the epistemological limits of traditional bag-of-words approaches. The dictionary-based lexical baseline, while computationally efficient, systematically overestimated the salience of economic themes because it remains blind to syntax, context, and polysemy. By indiscriminately capturing superficial buzzwords and administrative jargon, lexical proxies generate a noisy, inflated metric that can easily mislead downstream statistical analyses.

By contrast, the implementation of a Zero-Shot semantic pipeline introduces a more precise framework. The specific engineering choice to utilize a chunk-level segmentation coupled with a max-pooling aggregation strategy explicitly solves the pervasive problem of thematic dilution in long-form

political documents. This algorithm successfully filtered out performative political noise, retaining only substantive grounded argumentation. Crucially, the fact that both the naive lexical model and the advanced semantic model yielded the exact same near-zero correlation with the INSEE macroeconomic data acts as a powerful test of internal validity. This methodological convergence guarantees that the observed geographical disconnect is a sociological phenomenon, rather than a mere artifact of text-processing limitations.

6.3 Limitations of the Study

Despite the coherent findings generated by the dual NLP pipeline, several methodological limitations must be acknowledged, primarily stemming from data historicality and algorithmic anachronism. First, the reliance on 1982 INSEE localized statistics as a temporal proxy for the 1981 legislative elections introduces a slight asynchronous comparison. While this substitution is necessitated by historical data availability and remains statistically viable, it creates a temporal mismatch. The macroeconomic reality measured in 1982 may not perfectly reflect the exact localized economic anxiety experienced by candidates and voters during the spring 1981 campaign.

Second, the textual integrity of the *Archelec* corpus is inherently constrained by Optical Character Recognition (OCR) degradation. This issue is not uniform across the dataset; it is particularly pronounced in the older 1981 manifestos, which exhibit significantly higher noise levels and formatting irregularities than their 1993 counterparts. To maintain a high-quality input for the embedding models, a strict character-length filtering threshold was applied. While necessary to eliminate administrative artifacts, this mechanical filtering inevitably leads to the loss of extractable text. Consequently, poorly scanned but substantively relevant arguments may have been systematically discarded, marginally reducing the total discursive volume analyzed and introducing a slight survivorship bias into the textual data.

Finally, the deployment of advanced semantic models on historical text introduces the risk of semantic drift. The CamemBERT architecture is fundamentally trained on modern, contemporary French corpora. Consequently, projecting a 2020-era high-dimensional vector space onto 1980s political rhetoric creates an anachronistic gap. While the model excels in a Zero-Shot capacity for general semantic alignment, it may occasionally fail to capture historically situated political phrasing, archaic partisan idioms, or older industrial terminology. This implies that the semantic filter might conservatively underestimate certain nuanced expressions of the 1980s economic crisis that do not align seamlessly with contemporary linguistic patterns.

7 Conclusion

This study investigated the territorial anchoring of French political campaigns (1981–1993) by contrasting two NLP paradigms on the *Archelec* corpus against localized INSEE statistics. Methodologically, the results highlight the limits of bag-of-words techniques, which overestimate discursive intensity by ignoring syntactic context. Transitioning to a Zero-Shot semantic pipelineisolated substantive economic argumentation from superficial noise.

Most importantly, both pipelines revealed a persistent lack of spatial correlation between unemployment discourse and actual departmental unemployment rates. This disconnect provides robust quantitative evidence for the nationalization of French political communication. Addressing unemployment was a standardized rhetorical obligation across the political landscape, detached from local economic severity.

Future research could apply this methodology to recent elections to assess if modern campaigns remain nationalized, or extend the Zero-Shot framework to other salient themes, such as ecology or immigration, to further map the dynamics between objective demographics and political discourse.

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